Infant Deaths After Texas' 2021 Ban on Abortion in Early Pregnancy

Paper Replication

Gemmill et al. 2024 (JAMA Pediatrics) replicated by Michael Tu

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Overview, Background, and Research Question

- **Background:** Between 2018 and 2022 - there were over 100,000 infant mortalities in the United States, with over 10,000 of those mortalities occuring in Texas.
- **Setting:** In 2021 Texas passed SB8 - a law that prevented abortions past the first fetal heartbeat, with no exceptions for congenital anomalies.
- **Question:** The study looks to examine the counts and rates of infant mortality in the state, and determine whether or not the policy caused increased infant mortality.
- **Main Results:** Using a synthetic control design - the authors find that the state experienced **216 excess infant deaths** compared to the synthetic counterfactual
- **Replication contribution:** I used updated data (the original paper only had provisional data for some of its observation period) that includes more post-treatment time. I verify the effects that the original authors find, and find suggestions of persistent effects.

Data Overview

- The main data source for this analysis was **CDC WONDER**, a database of all deaths in the United States, with associated Multiple Cause of Death (MCD).
  - In addition, WONDER provides natality (birth) data, which we use to calculate mortality rates and provides variables for the synthetic control.
- The authors use data from 2018-2022; I confirm their original analyses and extend the observation period to 2024.

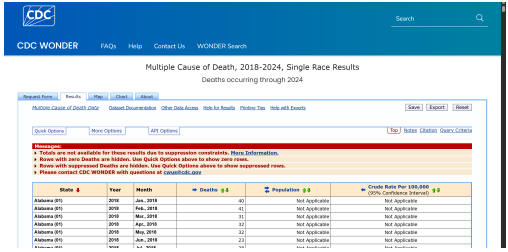


Figure: A screenshot of a query from the CDC Wonder databsae.

Methodology

- The paper's main result estimates the monthly differences in the number of observed vs counterfactual infant and neonatal deaths in Texas for January 2018 to December 2022.
  - An infant mortality is < 1 year of age, while neonatal is < 28 days.
- The counterfactual analysis is performed on the basis of a synthetic control trained on data from January 2018 to the end of February 2021.
  - The model is optimized on variables on the birth certificate: monthly births to those: 35 or older, college educated, or using Medicaid.
  - In addition, we use state fixed effects to account for unobserved state-level variation.

Summary Statistics

Table 1
Infant and Neonatal Deaths in Texas and the Rest of the US
2018-2024

	Texas				US Excluding Texas			
Year	No. of Deaths	Change from prior year, %	Rate	Change from prior year, %	No. of Deaths	Change from prior year, %	Rate	Change from prior year, %
Infant Deaths								
2018	2,083	--	5.50	--	18,607	--	5.45	--
2019	2,075	-0.38	5.50	-0.11	18,024	-3.13	5.35	-1.89
2020	1,968	-5.16	5.35	-2.73	16,728	-7.19	5.15	-3.63
2021	1,985	0.86	5.31	-0.60	17,136	2.44	5.21	1.03
2022	2,243	13.00	5.76	8.32	17,545	2.39	5.35	2.78
2023	2,259	0.71	5.82	1.18	17,109	-2.49	5.33	-0.36
2024	2,239	-0.89	5.73	-1.62	17,036	-0.43	5.26	-1.35
Neonatal deaths								
2018	1,382	--	3.65	--	12,024	--	3.52	--
2019	1,376	-0.43	3.64	-0.16	11,524	-4.16	3.42	-2.93
2020	1,300	-5.52	3.53	-3.11	10,626	-7.79	3.27	-4.26
2021	1,295	-0.38	3.47	-1.83	10,574	-0.49	3.21	-1.86
2022	1,432	10.58	3.67	6.00	10,771	1.86	3.29	2.26
2023	1,516	5.87	3.91	6.36	10,685	-0.80	3.33	1.36
2024	1,528	0.79	3.91	0.05	10,861	1.65	3.35	0.70

Notes: non-trivial differences in deaths outside of Texas; observe persistence in mortality in Texas after jump in 2022.

Weights

State	Gemmill et al. 2024		Replication	
	Weight for infant analysis	Weight for neonatal analysis	Weight for infant analysis	Weight for neonatal analysis
Alabama	16.6	--	21.8	--
Arizona	.7	--	4.0	--
Arkansas	-3.0	--	-3.4	--
California	-.6	7.6	4.7	16.7
Colorado	-1.3	--	-3.1	--
Florida	13.6	.1	16.0	9.5
Georgia	16.3	12.8	15.4	11.4
Illinois	13.4	23.5	5.5	12.4
Indiana	-8.1	-5.5	-9.1	-2.3
Kansas	1.4	--	-7.1	--
Kentucky	--	--	-.8	--
Louisiana	.7	7.1	7.7	19.7
Maryland	1.2	17.5	2.8	4.5
Massachusetts	10.1	--	--	--
Michigan	-1.1	-3.6	-1.7	-2.6
Minnesota	-1.3	--	-5.9	--
Mississippi	1.1	--	6.3	--
Missouri	34.2	8.8	40.2	22.1
New Jersey	1.2	.2	-1.1	-16.0
New York	-3.5	-4.7	4.1	-1.4
North Carolina	-5.5	-5.7	-4.1	-2.9
Ohio	.1	-2.8	-2.8	-3.3
Oklahoma	-2.2	--	.1	--
Pennsylvania	1.6	15.7	-2.3	11.0
South Carolina	-.3	.7	1.3	6.8
Tennessee	11.9	27.1	3.2	17.0
Virginia	-.1	-3.2	-3.6	-12.3
Washington	1.7	1.8	14.7	16.0
Wisconsin	1.3	2.7	-3.0	-6.4

Figure: Augmented Synthetic Control Weights

Notes: non-trivial differences in weights; although most top weighted states are consistent (Missouri, Florida, Georgia, Alabama). Kentucky added to analysis based off of inclusion criteria.

Main Result Pt.1

Estimated Difference Between Observed vs Expected Infant Deaths
Texas
March 2022 - December 2024

			Gemmill et al. 2024		Replication	
Month	Year	Observed Deaths	Estimated	Observed - Estimated (95% CI)	Estimated	Estimated - Observed (95% CI)
March	2022	184	169.5	14.5 (-11.2 to 40.2)	170.0	14 (-11.96, 39.97)
April	2022	200	163.6	36.4 (8.4 to 66.6)	165.5	34.48 (8.52, 60.45)
May	2022	163	169.3	-6.3 (-34.3 to 21.6)	171.0	-7.96 (-33.92, 18)
June	2022	193	165.6	27.4 (-2.8 to 57.5)	164.8	28.23 (2.27, 54.19)
July	2022	212	171.9	39.1 (6.7 to 76.0)	172.5	39.51 (13.54, 65.47)
August	2022	183	175.8	7.2 (-18.5 to 37.4)	177.4	5.62 (-20.34, 31.58)
September	2022	209	165.5	43.5 (15.5 to 71.4)	166.2	42.77 (16.81, 68.73)
October	2022	214	173.9	40.1 (7.7 to 72.5)	176.7	37.31 (11.34, 63.27)
November	2022	181	172.3	7.7 (-18.0 to 35.6)	171.8	9.19 (-16.77, 35.16)
December	2022	177	169.7	6.3 (-21.7 to 36.5)	168.7	8.27 (-17.69, 34.24)
January	2023	184			168.0	16.05 (-9.92, 42.01)
February	2023	152			167.4	-15.37 (-41.33, 10.6)
March	2023	196			172.6	23.36 (-2.6, 49.32)
April	2023	185			166.9	18.14 (-7.83, 44.1)
May	2023	197			169.7	27.26 (1.29, 53.22)
June	2023	209			173.6	35.38 (9.42, 61.34)
July	2023	213			171.4	41.57 (15.61, 67.54)
August	2023	196			172.7	23.3 (-2.66, 49.27)
September	2023	173			169.1	3.89 (-22.07, 29.85)
October	2023	171			162.7	8.34 (-17.62, 34.3)
November	2023	174			164.0	10.01 (-15.95, 35.97)
December	2023	209			166.4	42.58 (16.62, 68.54)
January	2024	157			176.7	-19.72 (-45.69, 6.24)
February	2024	162			167.4	-5.45 (-31.41, 20.52)
March	2024	192			175.0	17.01 (-8.96, 42.97)
April	2024	191			161.8	29.18 (3.22, 55.15)
May	2024	197			170.7	26.29 (0.33, 52.26)
June	2024	189			167.2	21.78 (-4.18, 47.74)
July	2024	183			172.2	10.8 (-15.16, 36.77)
August	2024	223			160.3	62.71 (36.75, 88.67)
September	2024	176			165.7	10.35 (-15.62, 36.31)
October	2024	202			161.6	40.43 (14.47, 66.39)
November	2024	187			172.1	14.95 (-11.02, 40.91)
December	2024	180			169.0	10.99 (-14.97, 36.95)
Total Difference (March 2022 - December 2022)			**218.9**		**211.4**	
Total Difference (March 2022 - December 2024)			**–**		**651.3**	

Main Result Pt.2

Estimated Difference Between Observed vs Expected Neonatal Deaths
Texas
March 2022 - December 2024

			Gemmill et al. 2024		Replication	
Month	Year	Observed Deaths	Estimated	Observed - Estimated (95% CI)	Estimated	Estimated - Observed (95% CI)
March	2022	118	107	11.0 (-12.8 to 34.9)	106.0	12 (-12.05, 36.06)
April	2022	135	107.7	27.3 (3.4 to 52.9)	107.4	27.56 (3.5, 51.62)
May	2022	97	107.8	-10.8 (-34.6 to 16.6)	111.1	-14.13 (-38.19, 9.92)
June	2022	128	106.7	21.3 (-6.1 to 46.9)	108.4	19.61 (-4.45, 43.66)
July	2022	140	105.8	33.2 (-3.0 to 62.4)	106.3	33.73 (9.67, 57.78)
August	2022	114	112.4	1.6 (-24.0 to 29.0)	114.2	-0.17 (-24.23, 23.89)
September	2022	131	106.9	24.1 (-1.5 to 51.5)	106.3	24.73 (0.68, 48.79)
October	2022	146	106.9	39.1 (13.4 to 62.9)	108.8	37.16 (13.1, 61.21)
November	2022	111	115.3	-5.3 (-31.0 to 23.8)	110.6	0.37 (-23.69, 24.42)
December	2022	115	111.8	3.2 (-20.6 to 28.8)	110.2	4.8 (-19.25, 28.86)
January	2023	125			111.9	13.07 (-10.99, 37.12)
February	2023	102			108.6	-6.56 (-30.61, 17.5)
March	2023	141			110.4	30.55 (6.5, 54.61)
April	2023	127			105.8	21.18 (-2.88, 45.23)
May	2023	123			109.7	13.33 (-10.72, 37.39)
June	2023	128			112.0	15.99 (-8.06, 40.05)
July	2023	152			104.4	47.56 (23.5, 71.61)
August	2023	136			110.5	25.51 (1.45, 49.56)
September	2023	114			108.5	5.51 (-18.55, 29.56)
October	2023	107			105.5	1.51 (-22.55, 25.56)
November	2023	122			102.7	19.27 (-4.78, 43.33)
December	2023	139			107.0	31.99 (7.94, 56.05)
January	2024	111			114.2	-3.18 (-27.24, 20.87)
February	2024	106			107.8	-1.75 (-25.81, 22.3)
March	2024	133			112.3	20.66 (-3.4, 44.71)
April	2024	129			103.9	25.1 (1.05, 49.16)
May	2024	140			111.7	28.27 (4.21, 52.32)
June	2024	122			117.7	4.29 (-19.77, 28.34)
July	2024	129			117.6	11.4 (-12.65, 35.46)
August	2024	147			106.0	40.99 (16.94, 65.05)
September	2024	124			107.7	16.3 (-7.76, 40.35)
October	2024	144			106.0	38.05 (13.99, 62.1)
November	2024	115			112.6	2.45 (-21.61, 26.5)
December	2024	128			105.8	22.22 (-1.84, 46.27)
Total Difference (March 2022 - December 2022)			**146.7**		**145.7**	
Total Difference (March 2022 - December 2024)			**--**		**557.3**	

Note: My conformal standard errors are calculated slightly differently, based on empirical distribution of pre-treatment gaps as opposed to inversion of the model.

Main Results Visualized Pt.1

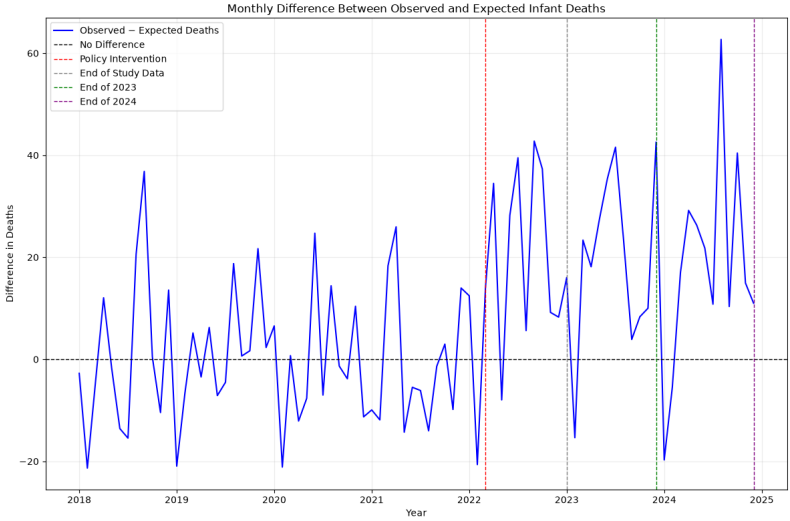


Figure: Monthly Differences in the Number of Observed vs Expected Infant Deaths

Main Results Visualized Pt.2

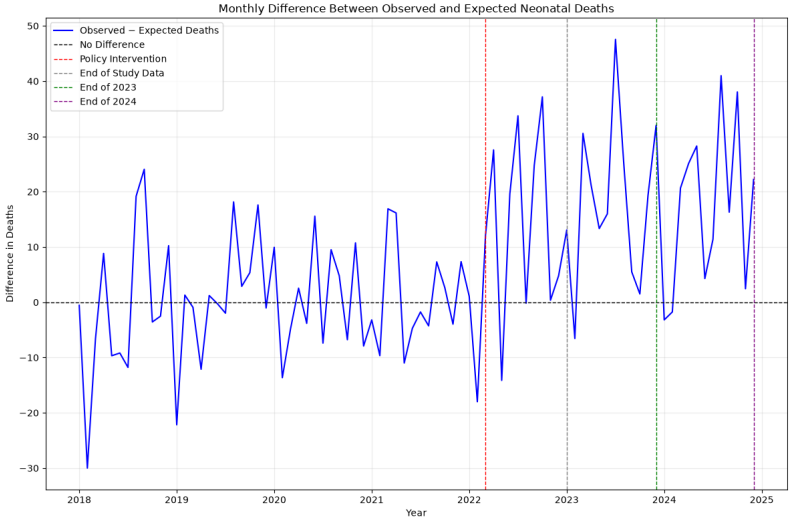


Figure: Monthly Differences in the Number of Observed vs Expected Neonatal Deaths

Discussion

- In the intervening years since data collection from the original paper to today, states have passed a number of laws restricting abortion.
  - It appears that neonatal mortality seems to be driving the overall mortality effect.
- Certain agents may have internalized this shock and substituted to other forms of reproductive care.
  - Shield laws
  - Travel to neighboring states
  - Medical Abortion
- The paper also includes some descriptive cause of death evidence that shows that congenital defects and complex pregnancy deaths among infants skyrocketed from 2021 to 2022 (while staying flat or negative for the rest of the US).
  - I replicate that result in the GitHub repository.

Conclusions and Next Steps for Final

- We are able to verify the paper's main results through two extra years of data
  - In addition, we are able to replicate the process using differently implemented software.
- The paper's synthetic design and analysis at the month-state level might be understating effects of the policy change.
- The next step I would like to pursue is leveraging border-effects, and staggered difference-in-difference designs, as other neighboring states have implemented similar abortion bans.

